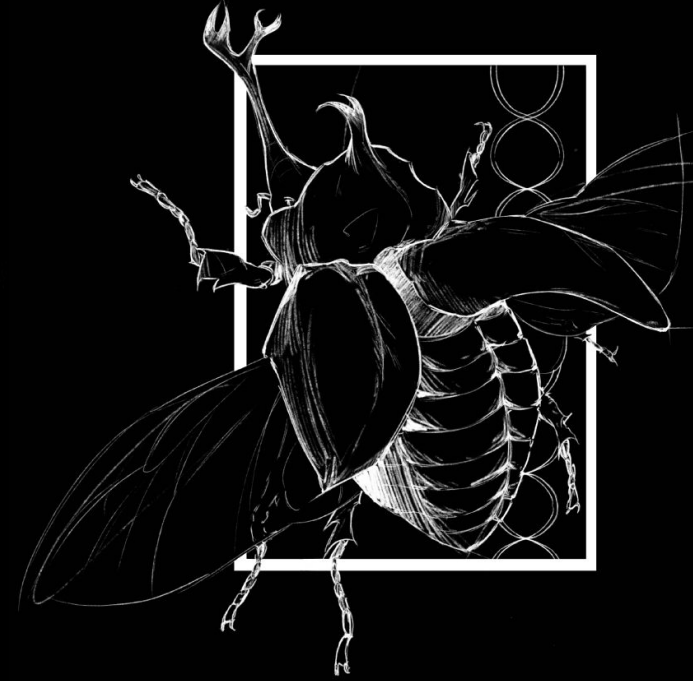




Background

How does effective population size modeled by population density and dispersal affect the rate of chromosome number change in Muridae? The primary goal of this study was to understand the differences in the rate of chromosome number evolution in the family Muridae. Muridae is the largest rodent family and encompasses more than 1000 species observed throughout the world today. Understanding the rates of chromosome number change allows us to see whether certain species are evolving at a faster rate. To study this vector of evolution through ecological data, effective population size was used as the variable of interest. The reasoning behind population size is tied to the drift barrier hypothesis first introduced by (insert citation), which states that small populations are more likely to experience weak selection. A weak selection allows for fission and fusion events to become fixed in a genetic pool, leading to more variable chromosome numbers. By utilizing phylogenetic data and comparative methodology that accounts for non-independent data, we are able to use data on living species to create estimates previous chromosome number transitions.

Study Design & AI

Haploid chromosome number data and a phylogenetic tree were obtained from the Blackmon lab. Population density and dispersal were used as proxies for population size and were collected from the COMBINE database. The data was then compiled and fit onto a phylogenetic tree that was pruned and converted into an ultrametric tree. Data was then standardised by converting the density and dispersal data into z-scores that were then added together to create a combined z-score. The median combined score was then calculated and the species scores above the median were classified as large populations while below were small populations. With a binary trait value of small/large population, the data is now ready to be fit to a Markov chain model, which uses current traits in living species to create probabilities for previous state changes in a phylogenetic tree. AI was used in this study to aid in the collection of data and refinement of hypotheses. Furthermore, AI was used to organize data in excel spreadsheets and also to aid with R-code script writing related to running the MCMC.

Model Fitting

A Markov Chain Model was used as the model of choice due to the necessity of chromePlus. The MCMC works by taking the state of small vs large population on the tips of an ultrametric tree, to estimate the likelihood of previous states, constructing a posterior distribution. This likelihood distribution for states can then be used to estimate the difference in rates of chromosome number evolution. ChromePlus was used because it is able to use MCMC to model fusion and fission which were our primary rates of interest for this study within a phylogenetic tree.



Results

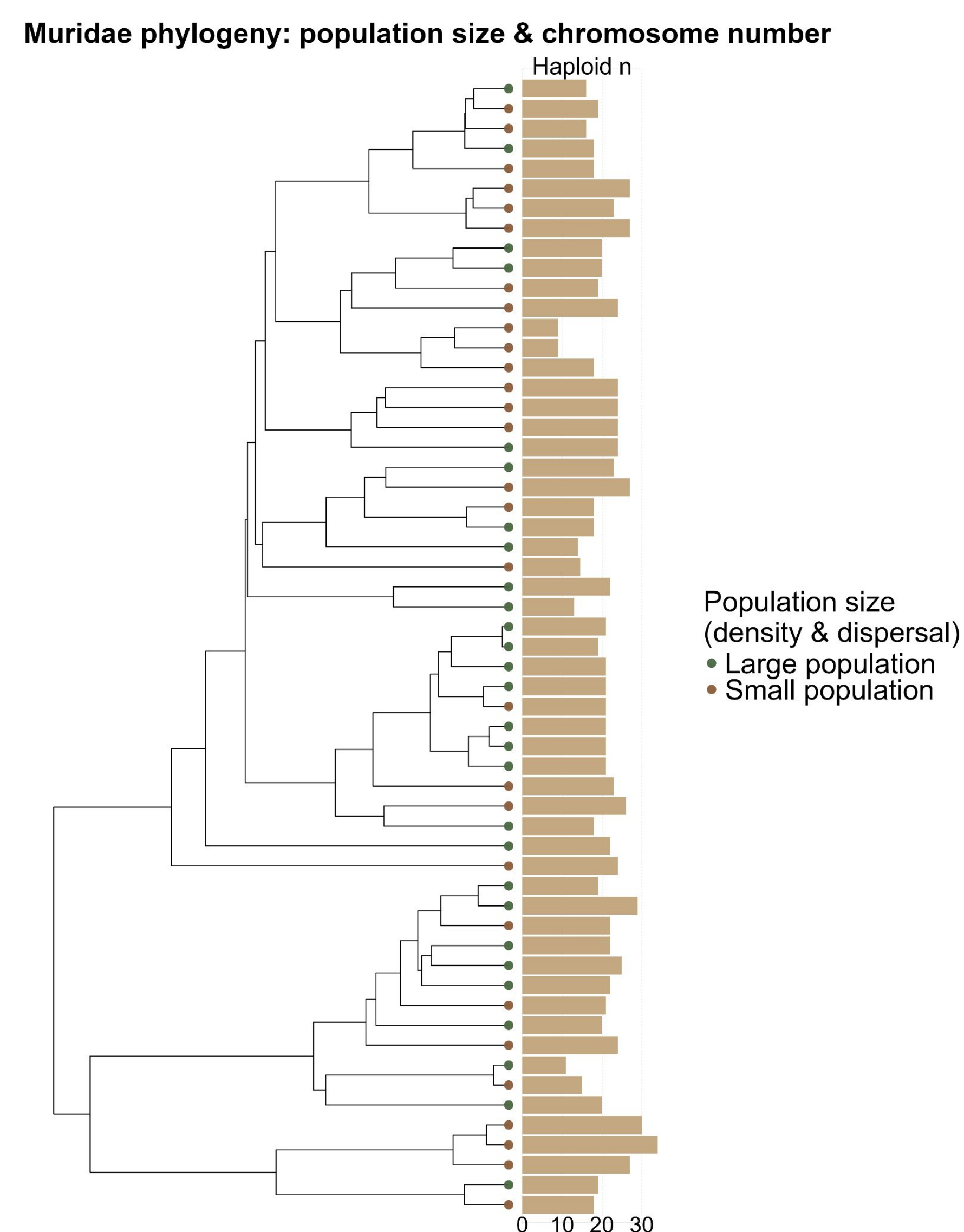


Figure 1. Phylogenetic tree with states of large population and small population. Haploid chromosome number is displayed on bar graph. Variability observed in both states and haploid chromosome number indicating suitability for analysis.

Rate differences (state 1 - state 2)

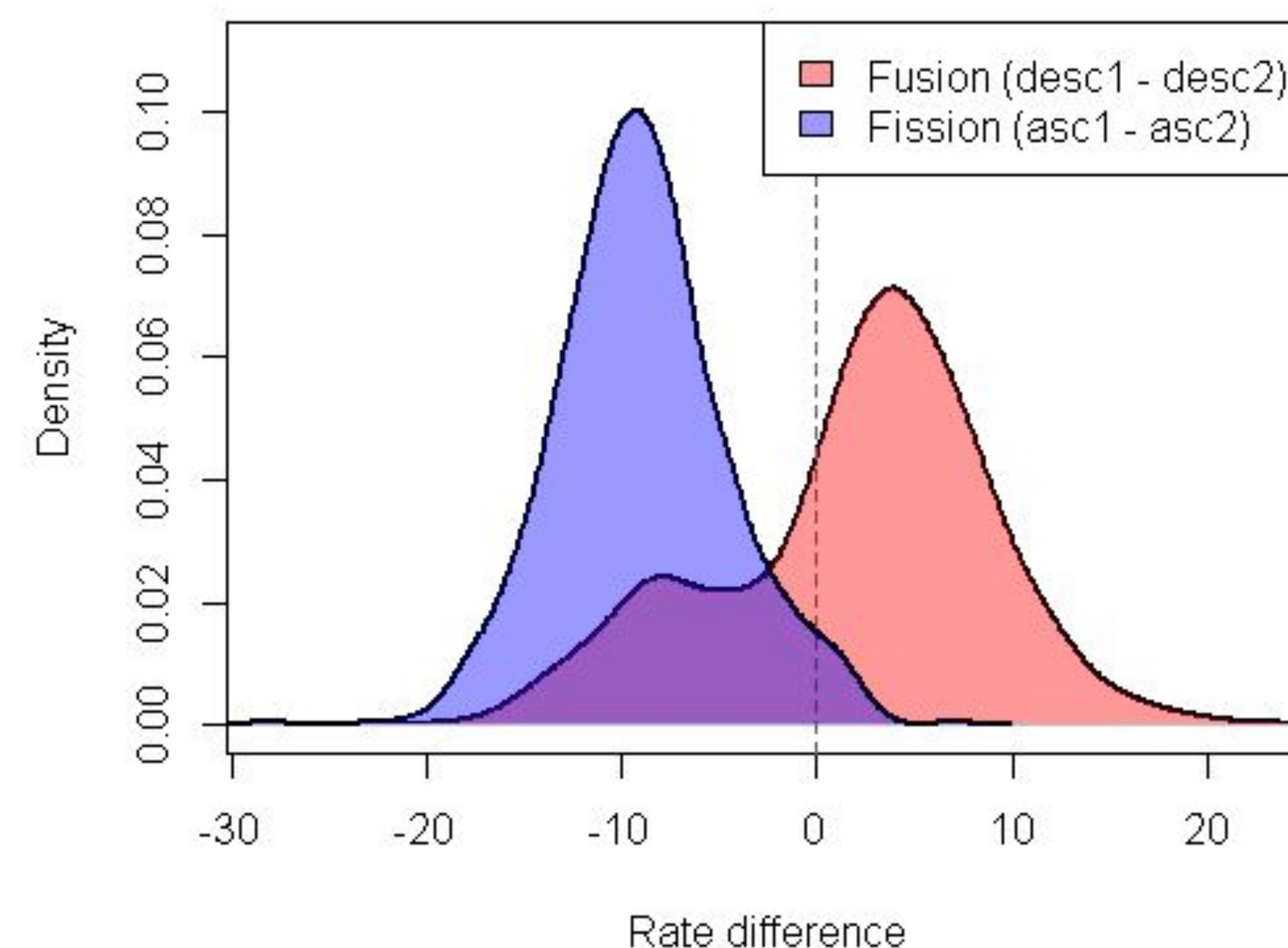


Figure 2. Rate Difference density chart of fusion (desc) and fission (gain) distributions modeled by chromePlus. Difference of small population state 1 - large population state 2 rates of fusion vs fission.

Discussion

My hypothesis was that smaller populations would experience higher rates of chromosome number evolution as a result of weak selection being overcome by genetic drift. Figure 1. shows a relevant biological variety of state expression helping to ensure no bias or inflation in the results. Figure 2 above displays a density chart showing a higher fission rate in large populations with a peak rate difference of -10. The fission distribution has 99.3% of the data below zero providing strong support for this finding. Figure 2 also shows smaller populations have a higher fusion rate with a peak rate difference just above the null rate difference. The fusion distribution has 88.8% of the data above zero providing weak support for this finding. Figure 2 also shows a wider distribution spread indicating a larger uncertainty in the results of fusion compared to fission. Thus, my results do not strongly support the hypothesis that chromosome number evolution is faster in small populations due to contrasting results in fission and fusion rates.

While the higher fusion rate supports the hypothesis, the contrasting results of higher rates of fission in large populations works against the hypothesis. However, both primary results do support the idea that population size has an effect on chromosome number evolution. One limitation of this study is that it doesn't directly provide an explanation for the difference in chromosome gain and loss events. The Asymmetric results could be due to a stronger selection of chromosome loss events in the small population samples and stronger drift fixation of fusion events present in the collected large population samples.

Furthermore, the Muridae family is extensive and this study is limited to the data collected on 57 species limiting its representative power. The separation of traits into small and large population is taken by the median of combined z-scores, however this distinction is a methodological simplification and should not be take as biologically grounded distinction. It should be recognized that sampling size means there is potential for intrinsic bias or inflation of posterior estimates. It should also be recognized that the population density and dispersal are limited proxies of effective population size and are not fully representative. The reasoning behind population density was the interpretation that a larger density would correlate with population size while population dispersal data would account for genetic drift via movement of genes between populations. The use of these proxies is something that should be considered in future studies with the hope that more proxies are utilized to create more biologically significant representations of population size. Future studies should endeavor to compile more data on population density and dispersal to build larger repositories of data that allow us to better understand the relationship between genetic drift and natural selection.

Conclusions & Acknowledgements

Results showed that large populations experience higher rates of fission which contrasts with our original hypothesis. This is something that requires future research to understand whether this is the result of an independent mechanism that works against the drift barrier hypothesis or a byproduct of limited sampling. Findings on fusion were uncertain possibly due to sample size and the model used.

This work was supported by Blackmon lab who provided the phylogenetic and chromosome data. Population data was obtained from COMBINE database.

Huge thanks to Dr. Blackmon and the CURES staff who made this possible and are working to make undergraduate resarch more available.